UNIVERSIDAD NACIONAL DE COLOMBIA

Applications of Semidefinite Programming in Quantum Information

Juan Manuel Segura Guatibonza
Caos & Complejidad - Universidad Nacional de Colombia
February 13, 2026.

Contents

Linear Programming

A linear program LP is an optimization problem which consists in maximising a linear function $f(x_1, \ldots, x_n)$, called the objective function, $f : \mathbb{R}^k \to \mathbb{R}$ such that it must satisfy some equality constraints $g_i : \mathbb{R}^k \to \mathbb{R}$ and inequality constraints $h_j : \mathbb{R}^k \to \mathbb{R}$ of the form of linear constraints as shown below:

$$\begin{aligned} g_i(\vec{x}) &= b_i \\ h_j(\vec{x}) &\leq c_j \end{aligned}$$

Linear Programming

Since linear functions can be codified as vectors, there exists vectors $\vec{a}$, $\vec{r_i}$ and $\vec{s_j}$ (where $i = 1, \ldots, m$ and $j = 1, \ldots n$) such that $f(\vec{x}) = \vec{a} \cdot \vec{x}$, $g_i(\vec{x}) = \vec{r_i} \cdot \vec{x}$ and $h_j(\vec{x}) = \vec{s_j} \cdot \vec{x}$.

Therefore, the LP may be expressed as follows:

$$\begin{array}{rl} \text{Maximise} & \vec{a} \cdot \vec{x} \\ \text{constrained to} & \vec{r_i} \cdot \vec{x} = b_i, \quad i = 1, \ldots, m \\ & \vec{s_j} \cdot \vec{x} \leq c_j, \quad i = 1, \ldots, m \end{array}$$

Linear Programming

Let B and C be the matrices of equality and inequality constraints respectively, and let $\vec{b}$ and $\vec{c}$ be the limits of the equality and inequality constraints respectively, such that

$$B = \begin{pmatrix} \vec{r}_1^T \\ \vdots \\ \vec{r}_m^T \end{pmatrix}, \quad C = \begin{pmatrix} \vec{s}_1^T \\ \vdots \\ \vec{s}_n^T \end{pmatrix}, \quad \vec{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}, \quad \vec{c} = \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix}.$$

Then the LP can be expressed as maximising $\vec{a} \cdot \vec{x}$ constrained to $B\vec{x} = \vec{b}$ and $C\vec{x} \leq \vec{c}$.

Linear Programming

A vector $\vec{x}$ which satisfies the constraints is a **feasible point**. The set of all feasible points is called the **feasible set** $\mathcal{F}$. This set is convex, which means that for $\vec{x}_1, \vec{x}_2 \in \mathcal{F}$ holds

$$\vec{y} = p\vec{x}_1 + (1-p)\vec{x}_2 \in \mathcal{F}, \quad \forall p \in [0,1] \,.$$

The **optimal value** α of the LP is it's solution and the **optimal variable** $\vec{x}^*$ is that which achieves the optimal value, that is:

$$\alpha = \max_{\vec{x}\in\mathcal{F}} \vec{a}\cdot\vec{x},$$

$$\vec{x}^* = \operatorname*{argmax}_{\vec{x}\in\mathcal{F}} \vec{a}\cdot\vec{x}.$$

Linear Programming

Let's consider as an example the l_1 and l_∞ norms given by

$$||\vec{y}||_1 = \sum_i |y_i|, \quad ||\vec{y}||_\infty = \max_i |y_i|.$$

This norms can be formulated as LP's in the following way:

$$\begin{aligned} ||\vec{y}||_1 = \text{minimise} \quad & \vec{1} \cdot \vec{x} \\ \text{constrained to} \quad & -\vec{x} \leq \vec{y} \leq \vec{x} \\ ||\vec{y}||_\infty = \text{minimise} \quad & x \\ \text{constrained to} \quad & -x\vec{1} \leq \vec{y} \leq x\vec{1} \end{aligned}$$

Linear Programming

```python
def l1_norm(y):
    n = len(y)
    # Optimization variable
    x = cp.Variable(n)
    # Vector of ones
    ones = np.ones(n)
    # Objective function
    objective = cp.Minimize(ones.T @ x)
    # Constraints
    constraints = [-x <= y,
                   y <= x]
    # Create and solve the problem
    prob = cp.Problem(objective, constraints)
    prob.solve()
    # Get the optimal value
    l1 = prob.value
    return l1
```

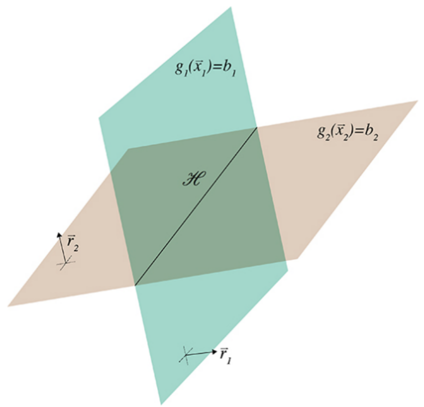
```python
def linf_norm(y):
    n = len(y)
    # Optimization variable
    x = cp.Variable()
    # Vector of ones
    ones = np.ones(n)
    # Objective function
    objective = cp.Minimize(x)
    # Constraints
    constraints = [-x * ones <= y,
                   y <= x * ones]
    # Create and solve the problem
    prob = cp.Problem(objective, constraints)
    prob.solve()
    # Get the optimal value
    linf = prob.value
    return linf
```

Linear Programming

If $\mathcal{F} = \emptyset$, then the linear program is **infeasible**. The subset of LP's which aim to check whether the feasible set is empty or not are known as **feasibility LP's**. The objective function in this LP's are constant $\vec{a} = 0$.

$$\alpha = \max_{\vec{x} \in \mathcal{F}} \vec{0} \cdot \vec{x} = \begin{cases} 0 & \text{if } \mathcal{F} \neq \emptyset \\ -\infty & \text{if } \mathcal{F} = \emptyset \end{cases}$$

Linear Programming

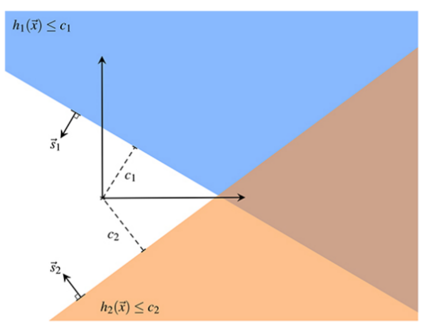


Equality constraints $\{g_i(\vec{x}) = b_i\}_i$ specify an affine subspace $\mathcal{H} \subseteq \mathbb{R}^k$ where $\vec{x}$ belongs to, that is, it defines and linear space which doesn't need to pass through the origin. If l is the number of linearly independent constraint equations, then the dimension of $\mathcal{H}$ is $k - l$.

Each equation $g_i(\vec{x}) = b_i$ specifies a hyperplane and the vector $\vec{r}_i$ which represents the linear function g_i is the normal vector to the hyperplane.

Linear Programming

Inequality constraints specify half-spaces, which divide the space into two, such that $\vec{x}$ relies in one half. The vectors $\vec{s}_j$ that represent the functions h_j are the normal vectors to the hyperplanes that divide the space. The hyperplanes are displaced from the origin in the direction $\vec{s}_j$ by the amount c_j.



The intersection between the half-spaces results in a **polyhedral set**, and if it is bounded it is called a **polytope**.

Linear Programming

The feasible set $\mathcal{F}$ is the intersection of the affine subspace $\mathcal{H}$ and the polyhedral set.

Every LP has an alternative formulation known as the **dual problem**, such that every feasible point it defines provides an upper bound to the **primal LP**. To each constraint of the primal LP there will be associated a Lagrange multiplier, known as **dual variable**. Let y_i and z_j be the dual variables.

$$\max \vec{a} \cdot \vec{x} \text{ constrained to } \qquad \begin{aligned} \vec{r}_i \cdot \vec{x} = b_i &\longrightarrow y_i \\ \vec{s}_j \cdot \vec{x} \leq c_j &\longrightarrow z_j \end{aligned}$$

Linear Programming

The **Lagrangian** of the problem is the following function:

$$\begin{aligned}\mathcal{L} &= \vec{a}\cdot\vec{x} + \sum_{i=1}^{m} y_i\left(b_i - \vec{r}_i\cdot\vec{x}\right) + \sum_{j=1}^{n} z_j\left(c_j - \vec{s}_j\cdot\vec{x}\right)\\ &= \left(\vec{a} - \sum_{i=1}^{m} y_i\vec{r}_i - \sum_{j=1}^{n} z_j\vec{s}_j\right)\cdot\vec{x} + \sum_{i=1}^{m} y_i b_i + \sum_{j=1}^{n} z_j c_j\end{aligned}$$

It is imposed that $z_j \geq 0$, $\forall j$. This implies that $\mathcal{L} \geq \vec{a}\cdot\vec{x}$, $\forall\vec{x}\in\mathcal{F}$.

Linear Programming

For the Lagrangian to be independent of the priml variable $\vec{x}$, it is required that the following constraint holds:

$$\vec{a} - \sum_{i=1}^{m} y_i \vec{r}_i - \sum_{j=1}^{n} z_j \vec{s}_j = 0.$$

Therefore, the Lagrangian is

$$\mathcal{L} = \sum_{i=1}^{m} y_i b_i + \sum_{j=1}^{n} z_j c_j.$$

Then, this Lagrangian is always bigger than the objective function evaluated in the feasible primal points.

Linear Programming

The tightest upper bound on the primal objective is obtained by minimising the Lagrangian over the dual variables, which is known as the **dual linear program** defined as follows:

$$\begin{aligned} \text{minimise} \quad & \mathcal{L} = \sum_{i=1}^{m} y_i b_i + \sum_{j=1}^{n} z_j c_j \\ \text{constrained to} \quad & \vec{a} - \sum_{i=1}^{m} y_i \vec{r}_i - \sum_{j=1}^{n} z_j \vec{s}_j = 0 \\ & z_j \geq 0, \quad j = 1, \ldots, n. \end{aligned}$$

Linear Programming

The primal problem optimises over k variables with m equalities and n inequalities, while the dual problem optimises over $n+m$ variables with k equalities and n inequalities.

The dual variables $(\vec{y}, \vec{z})$ that satisfy the constraints of the dual problem are **dual feasible**, which belong to the **dual feasible set** $\widehat{\mathcal{F}}$.

The **dual optimal value** is

$$\beta = \underset{(\vec{y},\vec{z})\in\widehat{\mathcal{F}}}{\text{minimise}}\ \vec{b}\cdot\vec{y} + \vec{c}\cdot\vec{z}.$$

Semidefinite Programming

A semidefinite program (SDP) is a constrained optimization problem where the variable is a Hermitian operator $X = X^\dagger$ which acts on a complex vector space. The **objective function** f is a real linear function in X, which can be represented by a Hermitian operator A such that $f(X) = \mathrm{tr}(AX)$. Then, the SDP is

$$\begin{aligned} \text{maximise} \quad & \mathrm{tr}(AX) \\ \text{constrained to} \quad & \Phi_i(X) = B_i, \quad i = 1, \ldots, m. \\ & \Gamma_j(X) \preceq C_j, \quad j = 1, \ldots, n. \end{aligned}$$

Φ_i and Γ_j are linear maps that are hermiticity-preserving.

Semidefinite Programming

An operator X which satisfies the constraints is **feasible**. The **feasible set** is the set of all feasible operators, and it is a convex set. The **optimal value** α and the **optimal variable** $\boldsymbol{X}^*$ are defined by

$$\alpha = \underset{X \in \mathcal{F}}{\mathrm{tr}(AX)} = \mathrm{tr}(AX^*)$$

If $\mathcal{F} = \emptyset$, then the SDP is said to be **infeasible**.

Semidefinite Programming

As an example, let's consider the problem of finding the maximum eigenvalue of a Hermitian operator H, which may be cast as the following SDP:

$$\begin{aligned} \text{maximise} \quad & \operatorname{tr}(H\rho) \\ \text{constrained to} \quad & \operatorname{tr}(\rho) = 1 \\ & \rho \succeq 0. \end{aligned}$$

Semidefinite Programming

Let Y_i be Lagrange multipliers or **dual variables** for the equality constraints and let Z_j be the dual variables for the inequality constraints. The **Lagrangian** is

$$\begin{aligned}\mathcal{L} &= \operatorname{tr}(AX) + \sum_{i=1}^{m} \operatorname{tr}\left\{Y_i\left[B_i - \Phi_i(X)\right]\right\} + \sum_{j=1}^{n} \operatorname{tr}\left\{Z_j\left[C_j - \Gamma_j(X)\right]\right\} \\ &= \operatorname{tr}\left\{X\left[A - \sum_{i=1}^{m}\Phi^{\dagger}(Y_i) - \sum_{j=1}^{n}\Gamma_j^{\dagger}(Z_j)\right]\right\} + \sum_{i=1}^{m}\operatorname{tr}(Y_iB_i) + \sum_{j=1}^{n}\operatorname{tr}(Z_jC_j)\end{aligned}$$

By demanding $Z_j \succeq 0$, $\forall j$, then $\mathcal{L} \geq \operatorname{tr}(AX)$, $\forall X \in \mathcal{F}$.

Semidefinite Programming

For the Lagrangian to be independent of the primal variable X it is required that

$$A - \sum_{i=1}^{m} \Phi^{\dagger}(Y_i) - \sum_{j=1}^{n} \Gamma_j^{\dagger}(Z_j) = 0.$$

Therefore,

$$\mathcal{L} = \sum_{i=1}^{m} \operatorname{tr}(Y_i B_i) + \sum_{j=1}^{n} \operatorname{tr}(Z_j C_j).$$

Semidefinite Programming

Since the Lagrangian is always bigger than the optimal value of the primal problem, then the **dual problem** consists in

$$\begin{aligned} \text{minimise} \quad & \mathcal{L} = \sum_{i=1}^{m} \operatorname{tr}(Y_i B_i) + \sum_{j=1}^{n} \operatorname{tr}(Z_j C_j) \\ \text{constrained to} \quad & A - \sum_{i=1}^{m} \Phi^\dagger(Y_i) - \sum_{j=1}^{n} \Gamma_j^\dagger(Z_j) = 0. \end{aligned}$$

$$\beta = \underset{(Y_1,\ldots,Y_m,Z_1,\ldots,Z_n)\in\widehat{\mathcal{F}}}{\text{minimise}} \mathcal{L} = \sum_{i=1}^{m} \operatorname{tr}(Y_i B_i) + \sum_{j=1}^{n} \operatorname{tr}(Z_j C_j)$$

Quantum States

Quantum states are described by density operators ρ, which are semidefinite operators with unitary trace, that is

$$\rho \succeq 0, \quad \mathrm{tr}(\rho) = 1.$$

Therefore, semidefinite programming results being useful in quantum information. The conditions for ρ being a density operator can be seen as SDP constraints.

Quantum States

Let's suppose there are given the measurement results of N observables $M_1, \ldots, M_N$, where the results obtained for each of the observables are their average values $\langle M_1 \rangle = m_1, \ldots, \langle M_N \rangle = m_N$. The **quantum state estimation** consists in finding a quantum state ρ such that it's expected values with respect to the measurements are the same as the ones from the data, that is:

$$\begin{aligned} \text{find} \quad & \rho \\ \text{constrained to} \quad & \operatorname{tr}(M_x \rho) = m_x \\ & \operatorname{tr}(\rho) = 1 \\ & \rho \succeq 0 \end{aligned}$$

Quantum States

There are three possible cases for this problem:

- Unique solution: The observables $\{M_x\}$ are tomographically complete. The M_x are linearly independent and $\mathcal{F} = \{\rho^*\}$.
- No solution: The SDP is infeasible and $\mathcal{F} = \emptyset$.
- Multiple solutions: There are infinite ρ that satisfy the constraints.

Quantum States

Let σ and ρ be quantum states where σ is a target state that is believed was prepared and produces the statistics. If the statistics are infeasible, it is of interest to measure how close is it the prepared state from the target. For this, the **trace distance** between states is defined as follows:

$$T(\rho, \sigma) = \frac{1}{2}||\rho - \sigma||_1.$$

The trace distance is a convex function of either state, that is

$$T(q\rho_0 + (1-q)\rho_1, \sigma) \leq qT(\rho_0, \sigma) + (1-q)T(\rho_1, \sigma).$$

Quantum States

The problem of finding the minimal distance or the closest state can be cast as follows:

$$
\begin{aligned}
\text{minimise} \quad & \frac{1}{2}\|\rho - \sigma\|_1, \\
\text{subject to} \quad & \operatorname{tr}(M_x \rho) = m_x, \\
& \operatorname{tr}(\rho) = 1, \\
& \rho \succeq 0.
\end{aligned}
$$

$$
\begin{aligned}
\text{minimise} \quad & \frac{1}{2}\operatorname{tr}(X), \\
\text{subject to} \quad & -X \succeq \rho - \sigma \succeq X, \\
& \operatorname{tr}(M_x \rho) = m_x, \\
& \operatorname{tr}(\rho) = 1, \\
& \rho \succeq 0.
\end{aligned}
$$

Quantum States

The **fidelity** between two mixed states ρ and σ is another notion for the closeness between those states, and it is defined as

$$\begin{aligned} F(\rho,\sigma) &= ||\sqrt{\rho}\sqrt{\sigma}||_1^2 \\ &= \left[\mathrm{tr}\left(\sqrt{\sqrt{\sigma}\rho\sqrt{\sigma}}\right)\right]^2 \end{aligned}$$

$$\begin{aligned} \sqrt{F(\rho,\sigma)} = \text{maximise} \quad & \mathrm{tr}(Y) \\ \text{constrained to} \quad & \begin{pmatrix} \rho & Y + iZ \\ Y - iZ & \sigma \end{pmatrix} \geq 0 \end{aligned}$$

Quantum States

However, if we do not have the state ρ but instead we have the expectation values $\{m_x\}$ of a set of measurements $\{M_x\}$ of the state, we may want to state the SDP such that it finds the state ρ which maximises the fidelity with respect to the target state and reproduces the obtained statistics, that is

$$\begin{aligned} \text{maximise} \quad & \operatorname{tr}(Y), \\ \text{constrained to} \quad & \begin{pmatrix} \rho & Y + iZ \\ Y - iZ & \sigma \end{pmatrix} \geq 0, \\ & \operatorname{tr}(M_x \rho) = m_x, \\ & \operatorname{tr}(\rho) = 1, \\ & \rho \succeq 0. \end{aligned}$$

Quantum States

Since it is impossible to obtain the expectation values exactly, but intstead the statistics known have some uncertainty, then for each measurement expectation values m_x there exists some uncertainty Δm_x such that the measurement is known to lie in the interval $[m_x - \Delta m_x, m_x + \Delta m_x]$.

$$
\begin{aligned}
\text{find} \quad & \rho, \\
\text{constrained to} \quad & m_x - \Delta m_x \leq \operatorname{tr}(M_x \rho) \leq m_x + \Delta m_x, \\
& \operatorname{tr}(\rho) = 1, \\
& \rho \succeq 0.
\end{aligned}
$$

Quantum States

For multipartite quantum states, the **quantum marginal problem** consists in determine the global state given some marginal states which are perfectly known.

$$
\begin{aligned}
\text{find} \quad & \sigma_{XYZ}, \\
\text{constrained to} \quad & \operatorname{tr}_Z(\sigma_{XYZ}) = \rho_{XY}, \\
& \operatorname{tr}_Y(\sigma_{XYZ}) = \rho_{XZ}, \\
& \operatorname{tr}_X(\sigma_{XYZ}) = \rho_{YZ}, \\
& \sigma_{XYZ} \succeq 0, \\
& \operatorname{tr}(\sigma_{XYZ}) = 1.
\end{aligned}
$$

Quantum States

By taking into account a target state ρ_{XYZ}, then it is possible to define a SDP which minimises the trace distance between this state and the variable σ_{XYZ} of the SDP.

$$\begin{aligned}
\text{minimise} \quad & \frac{1}{2}\operatorname{tr}(\Gamma_{XYZ}), \\
\text{constrained to} \quad & \Gamma_{XYZ} \succeq \sigma_{XYZ} - \rho_{XYZ}, \\
& \Gamma_{XYZ} \succeq -(\sigma_{XYZ} - \rho_{XYZ}), \\
& \operatorname{tr}_Z(\sigma_{XYZ}) = \rho_{XY}, \quad \operatorname{tr}_Y(\sigma_{XYZ}) = \rho_{XZ}, \quad \operatorname{tr}_X(\sigma_{XYZ}) = \rho_{YZ} \\
& \sigma_{XYZ} \succeq 0, \quad \operatorname{tr}(\sigma_{\mathrm{XYZ}}) = 1.
\end{aligned}$$

Quantum Measurements

Measurements in quantum mechanics are described by POVM elements of a measurement $\mathbb{M} = \{M_a\}$, where $M_a \succeq 0$ and $\sum_a M_a = \mathbb{I}$. Therefore, semidefinite programming is a useful tool for optimizing over the set of quantum measurements.

The **quantum measurement estimation problem** then consists of finding the set of measurements that reproduces experimental data of a given set of quantum states $\{\rho_x\}_x$. That is:

$$
\begin{aligned}
\text{Find} \quad & \mathbb{M} \\
\text{constrained to} \quad & \mathrm{tr}(M_a \rho_x) = p(a|x), \\
& M_a \succeq 0, \quad \sum_a M_a = \mathbb{I}.
\end{aligned}
$$

Quantum Measurements

There are three posibilities in this problem:

- $\{\rho_x\}_x$ is tomographically complete and the feasible set consists in a unique measurement $\mathcal{F} = \{\mathbb{M}^*\}$.
- The statistics are compatible but not tomographically complete. The feasible set consists in infinite measurements.
- The statistics are not compatible $\mathcal{F} = \emptyset$.

Quantum Measurements

This problem can be cast as the following SDP:

$$\begin{aligned} \text{Minimise} \quad & \delta \geq 0, \\ \text{constrained to} \quad & p(a|x) - \delta \leq \operatorname{tr}(M_a \rho_x) \leq p(a|x) + \delta, \\ & M_a \succeq 0, \\ & \sum_a M_a = \mathbb{I}. \end{aligned}$$

If $\delta^* = 0$, the quantum measurement estimation problem results being feasible.

Quantum Measurements

It can be shown that the dual formulation of this SDP is the following one:

$$
\begin{aligned}
\text{Maximise} \quad & \sum_{a,x} (v_{ax} - u_{ax}) p(a|x) + \operatorname{tr}(Y), \\
\text{constrained to} \quad & Y + \sum_{x} (v_{ax} - u_{ax}) \rho_x \preceq 0, \quad a = 1, \ldots, o, \\
& \sum_{a,x} (u_{ax} + v_{ax}) = 1, \\
& u_{ax} \geq 0, \quad v_{ax} \geq 0, \quad \forall a, x.
\end{aligned}
$$

Quantum Measurements

Let's suppose that there is a source of known quantum states where the outcomes follow a known probability distribution. The **quantum state discrimination** problem consists in identifying correctly the outcome state.

$$\text{Source} \longrightarrow \mathcal{E} = \{q_i, \rho_i\}$$

Therefore, the *quantum state discrimination* problem can be seen as codifying classical information in quantum states, where the information is codified in i according to it's distribution q_i. Then, the problem consists in finding the optimal measurement $\mathbb{M}$ such that it maximises the probability of guessing correctly.

Quantum Measurements

Given a state ρ_i, let's assume a measurement $\mathbb{M} = \{M_a\}_a$ is being performed. Then,

$$\begin{aligned} p(a|i) &= \operatorname{tr}(M_a \rho_i) \\ P_{guess} &= p(a = i) = \sum_i \operatorname{tr}(M_i \rho_i) \end{aligned}$$

Then, the following SDP can be cast:

$$\begin{aligned} P^*_{guess} = \text{Maximise} \quad & \sum_i q_i \operatorname{tr}(M_i \rho_i) \\ \text{constrained to} \quad & M_a \succeq 0, \quad \forall a, \quad \sum_a M_a = \mathbb{I}. \end{aligned}$$

Quantum Measurements

Now, let's consider that the measurement elements are fixed and the optimization muste be done with respect to the ensemble of quantum states $\mathcal{E} = \{q_i, \rho_i\}$.

$$\begin{aligned} P^*_{guess} = \text{Maximise} \quad & \sum_i q_i \operatorname{tr}(M_i \rho_i), \\ \text{constrained to} \quad & q_i \geq 0, \forall i, \quad \sum_i q_i = 1, \\ & \rho_i \succeq 0, \quad \operatorname{tr}(\rho_i) = 1, \forall i. \end{aligned}$$

This definition isn't good enough since it is maximised by a single state with unit probability and it doesn't capture properly the idea of an ensemble.

Quantum Measurements

By normalizing the probability of guessing the state correctly with respect to the probability of guessing correctly without making any measurement $\max_i q_i$, we obtain the following optimization problem:

$$\begin{array}{ll} \text{Maximize} & \dfrac{\sum_i q_i \operatorname{tr}(M_i \rho_i)}{\max_j q_j} \\ \text{subject to} & q_i \geq 0, \forall i, \quad \displaystyle\sum_i q_i = 1. \end{array}$$

This isn't a SDP since the objective function isn't linear. However, there exists an associated SDP that solves this problem.

Quantum Measurements

The following SDP solves the *quantum state discrimination* problem where the optimization is done with respect to the ensemble:

$$\begin{aligned} \text{Maximise} \quad & \sum_i \operatorname{tr}(M_i \omega_i) \\ \text{constrained to} \quad & \omega_i \succeq 0, \quad \operatorname{tr}(\omega_i) \leq 1, \forall i. \end{aligned}$$

Quantum Measurements

The Lagrangian associated to this formulation of the *quantum state discrimination* problem is the following:

$$\begin{aligned}\mathcal{L} &= \sum_i \mathrm{tr}(M_i\omega_i) + \sum_i \mathrm{tr}(Y_i\omega_i) + \sum_i z_i\left[1 - \mathrm{tr}(\omega_i)\right] \\ &= \sum_i \mathrm{tr}\left[\omega_i\left(M_i + Y_i - z_i\mathbb{I}\right)\right] + \sum_i z_i.\end{aligned}$$

Quantum Measurements

Therefore, the correspondent dual formulation of this SDP is the following:

$$\begin{aligned} \text{Minimise} \quad & \sum_i z_i \\ \text{constrained to} \quad & M_i + Y_i = z_i \mathbb{I}, \\ & Y_i \succeq 0, \quad z_i \geq 0, \forall i. \end{aligned}$$

Let $r = \sum_a z_a - 1$, then $N_a = \frac{1}{r} Y_a$ is such that $\mathbb{N} = \{N_a\}_a$ is a valid measurement.

$$p(a) = \frac{z_a}{1+r} : \quad \text{Normalized probability distribution.}$$

Quantum Measurements

Then, the dual formulation can be cast as:

$$\begin{aligned} \text{Minimise} \quad & 1 + r \\ \text{constrained to} \quad & \frac{M_a + r N_a}{1 + r} = p(a)\mathbb{I} \\ & N_a \succeq 0, \quad \sum_a N_a = \mathbb{I}, \\ & p(a) \geq 0, \quad \sum_a p(a) = 1, \quad r \geq 0. \end{aligned}$$

This quantifier is known as the *Generalized robustness of measurement informativeness*.

Quantum Entanglement

A quantum bipartite pure state $|\Psi\rangle_{AB}$ is **entangled** if it cannot be expressed as a tensor product between a state $|\psi\rangle_A \in \mathcal{H}_A$ and $|\phi\rangle_B \in \mathcal{H}_B$, that is:

$$|\Psi\rangle_{AB} \neq |\psi\rangle_A \otimes |\phi\rangle_B$$

For mixed bipartite systems, a quantum state ρ_{AB} is *entangled* if and only if cannot be written as a convex combination of product states:

$$\rho_{AB} \neq \sum_{\lambda} p_\lambda \rho_\lambda^A \otimes \rho_\lambda^B$$

The problem of determining if a state is entangled or not is a NP-hard problem in the sense that it doesn't exist an efficient algorithm that can determine this.

Quantum Entanglement

Let $X_{AB} \in \mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$ and let $\{|i\rangle\}$ be an orthonormal basis for $\mathcal{H}_B$. The **partial transpose** is defined as:

$$X_{AB}^{T_A} = \sum_{i,j} (|j\rangle\langle i| \otimes \mathbb{I}) X_{AB} (|j\rangle\langle i| \otimes \mathbb{I})$$

The *positive-partial transpose* criterion states that if the partial transpose of a quantum state ρ_{AB} isn't semidefinite positive, then ρ_{AB} is entangled, that is:

$$\rho_{AB}^{T_A} \not\succeq 0 \longrightarrow \rho_{AB} \neq \sum_{\lambda} p_\lambda \rho_\lambda^A \otimes \rho_\lambda^B$$

Quantum Entanglement

The set of entangled states can be classified between PPT and NPT according if they are positive or negative partial transpose respectively. Because there are entangled states that belong to the PPT set, it cannot be stablished anything about the separability of a PPT state. Let S_{PPT} be the set of positive partial transpose states, and S_{sep} the set of separable states (not entangled), then:

$$S_{sep} \subseteq S_{PPT}.$$

Quantum Entanglement

A measure of the "negativity" of an operator is the absolute value of the sum of it's negative eigenvalues. Let A be an hermitian operator, then:

$$A = A^{(+)} - A^{(-)},$$

where $A^{(+)}, A^{(-)} \succeq 0$. The positive eigenvalues and it's correspondent eigenprojectors form $A^{(+)}$ while the absolute value of the negative eigenvalues and their correspondent eigenprojectors form $A^{(-)}$. The *negativity* of A is defined as:

$$\mathcal{N}(A) = ||A^{-}||_1 = \sum_{\lambda<0} |\lambda|$$

Quantum Entanglement

Then, the **negativity of entanglement** can be cast as the following SDP (left side) and it's dual formulation (right side):

$$\begin{aligned} \mathcal{N}(\rho_{AB}) = \text{Minimise} \quad & \operatorname{tr}(Y), \\ \text{constrained to} \quad & \rho_{AB}^{T_A} = Z - Y, \\ & Y, Z \succeq 0. \end{aligned} \qquad \begin{aligned} \mathcal{N}(\rho_{AB}) = \text{Maximise} \quad & -\operatorname{tr}(X^{T_A}\rho_{AB}), \\ \text{constrained to} \quad & \mathbb{I} \succeq X, \\ & X \succeq 0. \end{aligned}$$

Quantum Entanglement

The operator W is known as an **entanglement witness**.

$$\begin{aligned} \mathcal{N}(\rho_{AB}) = \text{Maximise} \quad & -\operatorname{tr}(W\rho_{AB}) \\ \text{constrained to} \quad & \mathbb{I} \succeq W^{T_A}, \quad W^{T_A} \succeq 0. \end{aligned}$$

Let σ_{AB} be a separable state. Since $W^{T_A}, \sigma_{AB}^{T_A} \succeq 0$, it follows:

$$\operatorname{tr}(W^{T_A}\sigma_{AB}^{T_A}) = \operatorname{tr}(W\sigma_{AB}) \geq 0.$$

Therefore, if $\operatorname{tr}(W\rho_{AB}) \leq 0$, it follows that ρ_{AB} is entangled. However, *entanglement witnesses* are only able to detect NPT entangled states.

Quantum Entanglement

Another quantifier of entanglement is the **random robustness of entanglement**, which can be interpreted as the minimum amount of noise required to make a state into a separable one. It is defined as the following optimization problem:

$$
\begin{aligned}
R_R(\rho_{AB}) = \text{Minimise} \quad & r, \\
\text{constrained to} \quad & \frac{\rho_{AB} + r\frac{\mathbb{I}_A \otimes \mathbb{I}_B}{d_A d_B}}{1 + r} \in S_{sep}, \\
& r \geq 0.
\end{aligned}
$$

Since it doesn't exist an efficient algorithm in order to determine if a state is separable or not, it may be useful to relax this problem given some other criterion.

Quantum Entanglement

Given the the PPT criterion, the *random robustness of entanglement* can be relaxed into $R_R^{NPT}(\rho_{AB})$, which is the minimum amount of noise required to make a state into a PPT state, that is the following SDP:

$$
\begin{aligned}
R_R^{NPT}(\rho_{AB}) = \text{Minimise} \quad & r \\
\text{constrained to} \quad & \left(\rho_{AB} + r\frac{\mathbb{I}_A \otimes \mathbb{I}_B}{d_A d_B}\right)^{T_A} \succeq 0, \\
& r \geq 0.
\end{aligned}
$$

If $R_R^{NPT}(\rho_{AB}) > 0$, then ρ_{AB} is entangled. Since $S_{sep} \subseteq S_{PPT}$, the optimal value of this SDP lower bound the original problem $R_R^{NPT}(\rho_{AB}) \leq R_R(\rho_{AB})$.

Quantum Entanglement

In order to look out for better relaxations for the separability constraint such that it could be possible to get tighter lower bounds for the *random robustness of entanglement*, it is useful to introduce *k-symmetric extensions*.

A bipartite state $\rho_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B$ has **k-symmetric extension** if it exist a $(k+1)$-partite state $\rho_{AB_1 \ldots B_k}$ such that the reduced density operators $\rho_{AB_j} = \rho_{AB}$ for all $j = 1, \ldots, k$.

Theorem: A bipartite state $\rho_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B$ is separable if and only if it does have a *k-symmetric extension* for every k.

Quantum Entanglement

It should be also demanded that the *k-symmetric extensions* are symmetric:

$$(\mathbb{I}_a \otimes \Pi_{sym})\rho_{AB_1 \dots B_k}(\mathbb{I}_a \otimes \Pi_{sym}) = \rho_{AB_1 \dots B_k},$$

where Π_{sym} is the projector onto the symmetric subspace. This is, the state is invariant under permutation of any B-subsystem.

Quantum Entanglement

Given this *k-extendible* states, the *random robustness of entanglement* can be relaxed by considering the amount of noise required to make a state *k-extendible*, that is the following SDP:

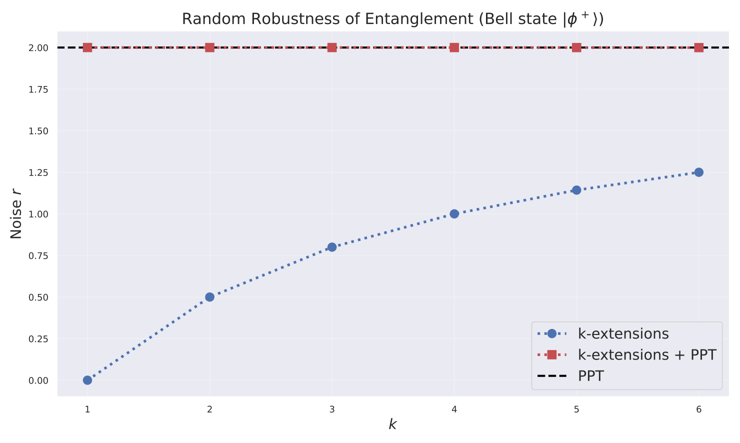
$$
\begin{aligned}
R_R^{kSE}(\rho_{AB}) = \text{Minimise} \quad & r \\
\text{constrained to} \quad & \rho_{AB} + r\frac{\mathbb{I}_A \otimes \mathbb{I}_B}{d_A d_B} = \tilde{\rho}_{AB_1} \\
& \tilde{\rho}_{AB_1 \dots B_k} \succeq 0, \quad r \geq 0, \\
& (\mathbb{I}_a \otimes \Pi_{sym})\tilde{\rho}_{AB_1 \dots B_k}(\mathbb{I}_a \otimes \Pi_{sym}) = \tilde{\rho}_{AB_1 \dots B_k}.
\end{aligned}
$$

Quantum Entanglement

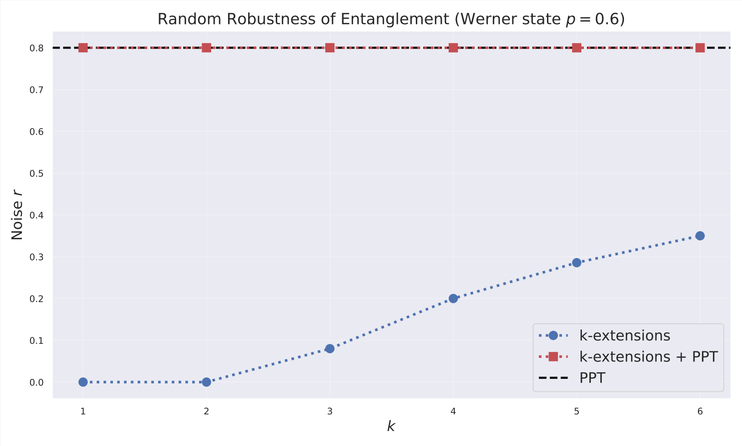
As k increases, this SDPs provide tighter lower bounds on the *random robustness of entanglement*, that is:

$$R_R^{1SE}(\rho_{AB}) \leq R_R^{2SE}(\rho_{AB}) \leq \cdots \leq \lim_{k\to\infty} R_R^{kSE}(\rho_{AB}) = R_R(\rho_{AB}).$$

Quantum Entanglement



Quantum Entanglement



Measurement Incompatibility

Can a set of measurements be performed simultaneously?

For projective measurements, the conmutator between the observables being measured provide a notion of compatibility between their correspondent measurements.

For generalized measurements POVM there exist a more general notion of incompatibility known as **non-joint measurability**.

Measurement Incompatibility

Let $\{\mathbb{M}_x\}_x$ be a set of m measurements where each of this measurements is specified by o elements of POVM, such that $\mathbb{M}_x = \{M_{a|x}\}_a$.

The set of measurements $\{\mathbb{M}_x\}_x$ is **compatible** if there exists a **parent measurement** $\mathbb{N} = \{N_\lambda\}$ and a conditional probability distribution $p(a|\lambda, x)$ such that the statistics of the individual measurements can be recovered given the outcome of the parent measurement, that is:

$$M_{a|x} = \sum_\lambda p(a|\lambda, x) N_\lambda$$

Measurement Incompatibility

Let $\vec{a} = (a_1, \ldots a_m)$ be the outcomes of the m individual measurements $\{\mathbb{M}_x\}_x$, $x = 1, \ldots, m$. Given the outcome λ of the parent measurement, the probability of obtaining the oucomes $\vec{a}$ is the following:

$$p(\vec{a}|\lambda) = \prod_{x=1}^{m} p(a_x|\lambda, x).$$

Then, the probability that the xth component is equal to a is:

$$p(a|\lambda, x) = \sum_{\vec{a}} \delta_{a_x, a} p(\vec{a}|\lambda).$$

Measurement Incompatibility

$$
\begin{aligned}
M_{a|x} &= \sum_{\lambda} p(a|\lambda, x) N_{\lambda}, \\
&= \sum_{\lambda} \sum_{\vec{a}} \delta_{a_x, a} p(\vec{a}|\lambda) N_{\lambda}, \\
&= \sum_{\vec{a}} \delta_{a_x, a} N'_{\vec{a}},
\end{aligned}
$$

where

$$
N'_{\vec{a}} = \sum_{\lambda} p(\vec{a}|\lambda) N_{\lambda} \quad , \quad \sum_{\vec{a}} N'_{\vec{a}} = \mathbb{I}
$$

Measurement Incompatibility

Therefore, $\mathbb{N}' = \{N'_{\vec{a}}\}_{\vec{a}}$ is a valid measurement. This measurement is known as **canonical parent measurement**, which has o^m outcomes. The *join-measurability* problem can be cast as the following feasability SDP:

$$
\begin{aligned}
\text{Find} \quad & \mathbb{N}, \\
\text{constrained to} \quad & M_{a|x} = \sum_{\vec{a}} \delta_{a_x, a} N_{\vec{a}}, \\
& N_{\vec{a}} \succeq 0, \\
& \sum_{\vec{a}} N_{\vec{a}} = \mathbb{I}.
\end{aligned}
$$

Measurement Incompatibility

In order to quantify the incompatibility between the set of measurements, the **random robustness of measurement incompatibility** is defined as the noise required to make the set of measurements *jointly-measurable*:

$$
\begin{aligned}
\text{Minimse} \quad & r \\
\text{constrained to} \quad & \frac{M_{a|x} + r\frac{\mathbb{I}}{o}}{1+r} = \sum_{\vec{a}} \delta_{a_x,a} N_{\vec{a}}, \quad \forall a, x, \\
& N_{\vec{a}} \succeq 0, \quad \forall \vec{a}, \\
& \sum_{\vec{a}} N_{\vec{a}} = \mathbb{I}, \quad r \geq 0.
\end{aligned}
$$

Measurement Incompatibility

This problem can be cast as the following SDP:

$$
\begin{aligned}
\text{Minimise} \quad & r \\
\text{constrained to} \quad & M_{a|x} + r\frac{\mathbb{I}}{o} = \sum_{\vec{a}} \delta_{a_x,a} N_{\vec{a}}, \quad \forall a, x, \\
& N_{\vec{a}} \succeq, \quad \forall \vec{a}, \\
& \sum_{\vec{a}} N_{\vec{a}} = (1+r)\mathbb{I}, \quad r \geq 0.
\end{aligned}
$$

Bibliography

- Skrzypczyk, P., & Cavalcanti, D. (2023). Semidefinite programming in quantum information science. IOP Publishing.
- Boyd, S. (2004). Convex Optimization. Cambridge University Press.
- Watrous, J. (2020). Quantum Information Theory [Lecture notes]. University of Waterloo. https://cs.uwaterloo.ca/ watrous/QIT-notes/